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Antihyperlipidemic effect of iridoid glycoside deacetylasperulosidic acid isolated from the seeds of *Spermacoce hispida* L. - A traditional antiobesity herb

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ABSTRACT

Ethnobotanical relevance: The interest on herbal health supplements for obesity is increasing globally. Our previous ethnobotanical survey in Tiruvallur district, Tamil Nadu, India indicated the use of *Spermacoce hispida* L. seeds for the treatment of obesity.

Aim of the study: This study was aimed to validate the traditional claim and to identify the antihyperlipidemic principle in the seeds of *Spermacoce hispida* using bioassay guided fractionation method.

Methods: Bioassay monitored fractionation of the aqueous extract from *Spermacoce hispida* seeds was carried out using triton WR 1339 induced hyperlipidemic animals. It yielded deacetylasperulosidic acid (DAA) as the active ingredient. Pharmacokinetic properties of DAA were predicted using DataWarrior and SwissADME tools. *In vitro* antiobesity and antihyperlipidemic effects of DAA were evaluated in 3T3L1 preadipocytes and HepG2 cells, respectively. The chronic antihyperlipidemic efficacy of DAA was evaluated in high fat diet fed rats.

Results: DAA did not show any mutagenic and tumorigenic properties. It bound with PPARα with comparable ligand efficiency as fenofibrate. The treatment with DAA significantly lowered the proliferation of matured adipocytes, but not preadipocytes. The treatment of steatotic HepG2 cells with DAA significantly decreased the LDH leakage by 43.03% ($P < 0.05$) at 50 μM concentration. In triton WR 1339 induced hyperlipidemic animals, the treatment with 50 mg/kg dose significantly lowered the TC, TG and LDL-c levels by 40.27, 46.00 and 63.65% respectively. In HFD fed animals, the treatment at 10 mg/kg decreased BMI and AC/TC ratio without altering SRBG. It also improved serum lipid, transaminases and phosphatases levels of HFD fed animals. The treatment lowered adipocyte hypertrophy and steatosis of hepatocytes.

Conclusion: This preliminary report supported the traditional use of *Spermacoce hispida* for the treatment of obesity. Further detailed investigations on the long term safety, efficacy and molecular mode of action of *Spermacoce hispida* and DAA will throw more light on their usefulness for the management of obesity.

1. Introduction

Obesity is one of the major global health issues; it is a known risk factor for various illnesses like type 2 diabetes, cardiovascular diseases, locomotor disabilities and certain cancers (Fruh, 2017). Prevalence of obesity is increasing worldwide; in 2016, about 650 million adults and 41 million children were obese worldwide (WHO, 2011). The socioeconomic changes in developing countries increased the prevalence of obesity as in

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developed countries (Bhurosy and Jeewon, 2014). Excess truncal and abdominal adiposity among South Asians compared to other races make them prone to various obesity related complications, even at lower BMI levels (Gulati and Misra, 2017). According to the ICMR-INDIAB survey, 135 and 153 million people in India are having generalized and abdominal obesity, respectively (Pradeepa et al., 2015). The prevalence of overweight and obesity among children in India is also increasing steadily from 2002 to 2012 (Vohra et al., 2011). It was estimated that obesity caused 2.8% loss in global gross domestic product in 2014 and studies indicated that there was a stable increase in the economic burden for the management of obesity (Tremmel et al., 2017).

Lifestyle modifications are considered as the first line intervention for the management of obesity though it is difficult to achieve. Various pharmacotherapies are available for the management of obesity and Orlistat is the only FDA approved medication for long term management (Pilitsi et al., 2018). Though bariatric surgery is considered as an effective therapy; it cannot be considered as a viable option for the better management of obesity in majority of the obese subjects. The use of alternative therapies for the management of obesity is also increasing steadily; according to a survey approximately 1/3rd of US population is using dietary supplements for weight reduction (Pillitteri et al., 2008). There is an increasing interest on nutraceuticals like *Garcinia cambogia, Hoodia gordonii,* green coffee extract, for the management of obesity (Ríos-Hoyo and Gutiérrez-Salmeán, 2016).

The genus *Spermacoce* is a member of Rubiaceae family and has 275 species worldwide (Raju et al., 2017). *Spermacoce hispida* L. (Synonym: *Borreria hispida* (L.) Schum.) was reported from India, China and Malaysia. It occurs as a ubiquitous weed up to 1200 m with elliptic-ovate leaves. The flowers are in axillary fascicles with pinkish white corolla and oblong, membranous seeds which resemble coffee beans (Matthew, 1995). In India, the decoction of the roasted seeds had been used as a substitute for coffee (Nadkarni and Nadkarni, 1976). The seeds are considered as tonic, alterative and stimulant. Our recent ethnobiological survey in Tiruvallur district, Tamil Nadu, India about the medicinal plants to treat cardiometabolic diseases indicated that the seeds of *Spermacoce hispida* as one of the highly prescribed plant drugs for the treatment of obesity (Esakkimuthu et al., 2016). It has also shown to have hepatoprotective, antioxidant and anti-inflammatory effects (Conserva and Jesu Costa Ferreira, 2012). Previous works indicated the presence of β - sitosterol, ursolic acid (Kapoor et al., 1969), D-mannitol (Subramanian and Nair, 1971), isorhmnatin (Purushothaman and Kalyani, 1979) and dalspinin (Sundaram and Vasanthi, 2019) in this species. This study is aimed to identify the bioactive ingredient in the aqueous extract of *Spermacoce hispida* seeds by bioassay guided fractionation.

2. Methodology

2.1. Chemistry

2.1.1. Chemicals & instruments

Solvents were purchased from Avra chemicals and they were glass distilled before use. Oven-dried glasswares were used for all the procedures. FT-IR spectrum was recorded on a PerkinElmer FT-IR grating spectrophotometer (Spectrum Two) in KBr disc. Specific rotation was taken on a Jasco digital polarimeter. ^{1}H and ^{13}C NMR spectra were measured on a Bruker Avance-700 (^{1}H NMR; 700 MHz, ^{13}C NMR; 175 MHz) with CD_3OD and TMS as an internal standard. Chemical shift values are given in δ-scale. HRESIMS data were obtained on a Q-Tof micro mass spectrometer (Micromass/Waters). Analytical HPLC was performed on Hitachi ELITE LaChrom system and preparatory HPLC was performed on JASCO Gulliver system. HPLC was performed with a Xterra MS C_{18} column (4.6 mm i.d. x 100 mm for analysis, 19 mm i.d. x 100 mm for preparative; Waters) packed with 5 µm of ODS eluted with a MeOH–H_2O gradient solvent system (0–5% MeOH; 20 min) and with 254 nm detection.

2.1.2. Plant materials

The seeds of *Spermacoce hispida* were collected from Kancheepuram district of Tamil Nadu, India during April, 2017 and the botanical identity was confirmed by PP, one of the authors of this communication. A voucher specimen (ES/EP/01) was deposited at the herbarium of Entomology Research Institute, Loyola College, Chennai for future reference.

2.1.3. Extraction and bioassay guided isolation of the active principle

One kilogram of the cleaned and washed seeds of *Spermacoce hispida* was soaked in 3 L of distilled water, under continuous stirring (150 rpm) at 70 °C for 12 h. The aqueous extract was filtered through Whatman No. 3 filter paper under vacuum and the filtrate was concentrated under reduced pressure. This extract (yield: 58 g) was then extracted with ethylacetate (100 mL x 3) (yield: 10 g) and methanol (100 mL x 2) (yield: 14 g). Bioactive methanol soluble portion (10 g) was packed in a flash column (Buchi) with silica gel (Avra, 300 mesh) in chloroform and eluted with the mixtures of chloroform and methanol with increasing amounts of methanol. The fraction eluted with 10% methanol (yield: 3.2 g) showed activity. The bioactive fraction (1.5 g) was subjected to preparative HPLC (JASCO Gulliver, Cosmosil XTerra prep MS C_{18}, 19 x 100 mm, Gradient H_2O (A) – MeOH (B) (0–80% B), 20 min, UV: 254 nm), when pure compound deacetylasperulosidic acid (DAA) (purity: 98.6%, yield: 0.25 g; 0.05% on dry wt. basis) was obtained.

2.2. In silico studies

2.2.1. ADME properties and molecular docking of DAA with PPARα

The pharmacokinetic properties of DAA were evaluated using DataWarrior software (Version 4.7.2) (Sander et al., 2015) and SwissADME web tool (Darvin et al., 2018). DAA was docked with PPARα (PDB ID: 117G) protein using Autodock tools (Schüttelkopf and Van Aalten, 2004; Dundas et al., 2006; Sanner, 1999; Morris et al., 2009). Detailed protocol is given as Supplementary data 1.

2.3. Bioassays

2.3.1. Chemicals and reagents

DMEM, FBS and trypsin were purchased from Gibco (India); triton WR1339, fenofibrate, dexamethasone, HEPES, IBMX, MTT, ORO, sodium palmitate and sodium oleate were purchased from Sigma Aldrich (Bangalore, India). Insulin, Antibiotic-antimycotic mixture, gentamycin and LDH assay kit were obtained from Hi-Media chemicals (Mumbai, India). Kits used for biochemical assessment were purchased from Agappe diagnostics (Kochi, India). All other chemicals and solvents of analytical grade were purchased from Qualigens.

2.3.2. Effect of DAA on the viability of 3T3L1 adipocytes

3T3-L1 preadipocyte cells were purchased from National Centre for Cell Sciences (Pune, India) and cultured in a 5% CO_2 environment at 37 °C in DMEM with 10% FBS, 2 mM glutamine, 20 mM HEPES, 50 U/ml penicillin, and 50 mg/ml streptomycin sulphate. For differentiation, DMEM containing 10% FBS, 0.5 mM IBMX, 2 mM dexamethasone and 1.7 mM insulin was used. For all the assays, the test materials were dissolved in DMSO and the level of DMSO was kept below 0.05% to the final volume of the media. In the first set of experiments, preadipocytes were seeded in 96 well plate at a density of 1×10^4 cells/well; the cells were allowed to adhere overnight. On the next day, the cells in treatment wells were treated with DAA for 24 h and the cells in control wells were treated with the vehicle. The viability of the cells was assessed using MTT assay in accordance with the method of Saravanan et al. (2014). In another set of experiments, preadipocytes were seeded in a 96 well plate at a density of 1×10^4 cells, left overnight and incubated in a differentiation medium for 48 h. The differentiated cells in treatment wells were treated with DAA for 24 h and the viability of the cells was assessed using MTT assay. The results were expressed as the percentage of cell growth compared with the control.

2.3.3. Effect of DAA on lipid accumulation in palmitate – oleate induced steatotic HepG2 cells

HepG2 cells used in this study were obtained from National Center for Cell Sciences, Pune (India). Effect of DAA on the viability of HepG2 cells was assessed using MTT assay and the dose range for efficacy evaluation was kept below its GI_{25} level. Steatosis was induced by administering palmitate and oleate to HepG2 cells at 1 mM concentration, since it resembles the circulating free fatty acid levels in NAFLD (Belfort et al., 2006; Hetherington et al., 2016). Culture conditions and study protocols were in accordance with our previously published work (Toppo et al., 2017). Briefly the cells were made steatotic by treating them with palmitate – oleate mixture with or without test materials and 24 h after treatment intracellular triglyceride content and the leakage of enzymes (ALT, AST and LDH) were assessed. The results of enzyme leakage were expressed as percent enzyme release according to the following formula

$$\%\ \text{enzyme release} = \left(\frac{\text{Enzyme level in the supernatant}}{\text{Enzyme level in the supernatant} + \text{Enzyme level in the lysate}}\right) \times 100$$

2.3.4. Animals

Wistar male rats purchased from King Institute of Preventive Medicine, Chennai were used in this study. The animals were maintained in polypropylene cages at 23 ± 2 °C with the relative humidity of 50 ± 5%. The animals were given feed and water *ad libitum* except in fasting conditions. Fasting was done between 06.00 a.m. and 10.00 a.m., if needed. The animal protocols were approved by the Institutional Animal Ethics Committee (IAEC/ERI/LC-02/17).

2.3.5. Test samples

The test samples were dissolved in a vehicle containing 0.9% NaCl, 0.5% CMC, 0.2% Tween-80 and 98.4% distilled water. Fenofibrate (50 mg/kg) was chosen as positive control and the vehicle alone served as negative control. The treatments were given once daily by gavage.

2.3.6. Evaluation of the antihyperlipidemic effect of the test materials using triton WR 1339 induced acute hyperlipidemia

The antihyperlipidemic efficacy of the test materials were assessed at the synthesis phase of acute hyperlipidemia induced by triton WR 1339 (Vogel and Vogel, 1997). After a physiological fasting, the animals were given an intraperitoneal injection of triton WR 1339 in PBS and the treatment was given orally immediately after induction. Thirty minutes after treatment, the animals were fed with standard pellet diet. The animals were euthanized under mild CO_2 anesthesia after 18 h of treatment; blood samples were collected by cardiac puncture and serum lipid profile was analysed on the same day of blood collection.

2.3.7. Evaluation of the antihyperlipidemic effect of the active compound, DAA in high fat diet fed animals

Hyperlipidemia was induced by feeding the animals with high fat diet (HFD). Twenty four rats (126.5 ± 5.2 g) were supplemented with HFD (Supplementary Table 1) for two months before the start of the experiment and throughout the study period. After two months of HFD feeding, the animals were randomized into four groups containing six animals in each group. The animals in Group I consisted of HFD fed animals treated with the vehicle. The animals in Group II were HFD fed animals treated with fenofibrate. The animals in groups III and IV were HFD fed animals treated with DAA at 5 and 10 mg/kg, respectively. The treatment was given for a month. At the end of the study period, the animals were euthanized under mild CO_2 anesthesia and the blood was collected by cardiac puncture. Blood was centrifuged at 3500 rpm for 15 min; serum was detached and stored at 4 °C. Liver and reteroperitoneal adipose tissues were fixed in 10% buffered formalin with PBS for histopathological analyses.

2.3.8. Biochemical estimations

The feed consumption and body masses were taken daily at 9.00 a.m. The masses of the liver, spleen, kidney and retroperitoneal adipose tissue were taken and expressed as percent body mass. Serum AST, ALT, ACP, ALP, LDH and HDL-c levels were measured using commercial spectrophotometric kits, on the same day of euthanasia. TC, TG, albumin and bilirubin levels were assessed on the next day using commercial kits. The level of LDL-c was calculated using Friedewald equation (Friedewald et al., 1972). The degree of liver damage by HFD feeding was scored in accordance with the previously published methodology (Itagaki et al., 2013). The parameters such as hepatocyte ballooning (zero for no ballooning and two for many ballooning), lobular inflammation (zero for no inflammation and three for inflammation at > 4 foci at 200x field) and steatosis (zero for < 5% and three for > 66%) were considered to evaluate the histological sections.

2.4. Statistics

Statistical analyses were done using Graphpad Prism (version 5) software. All the data were presented as mean ± SD. GI_{50} and GI_{25} values were calculated using linear regression analysis. Homogeneity of variances in the data was evaluated using Bartlett's test. Statistical significance was assessed by one way ANOVA and Tukey's HSD. The $P \leq 0.05$ was set as significance level for all data.

3. Results

3.1. Phytochemistry

Bioassay guided isolation of the aqueous extract of *Spermacoce hispida* seeds yielded deacetylasperulosidic acid (DAA) as the active antihyperlipidemic constituent. It was isolated as off-white amorphous powder and it was analysed for $C_{16}H_{22}O_{11}$. $[\alpha]_D = +8.5°$ [c, 0.4 MeOH]; UV(MeOH) λ_{max} 201 and 239 nm; IR ν_{Max}^{KBr} (cm^{-1}): 3368 (OH), 2927, 1737 (C=O), 1634 (C=C); ^1H NMR [??, CD_3OD, 700 MHz] 5.05 (1H, d, J = 9.00 Hz, H1), 7.63 (1H, s, H3), 3.01 (1H, t, J = 6.60 Hz, H5), 4.83 (1H, brd, J = 6.14 Hz, H6), 6.02 (1H, d, J = 2.55 Hz, H7), 2.56 (1H, t, J = 8.13 Hz, H9), 4.22, 4.46 (each 1H, d, J = 15.53 Hz, H10), 4.72 (1H, d, J = 7.90 Hz, H1'), 3.23–3.28 (3H, m, H2', H-4' and H-5'), 3.38 (1H, t, J = 8.90 Hz, H3'), 3.63 (1H, dd, J = 12.10 and 7.0 Hz, H6'a), 3.85 (1H, dd, J = 12.10 and 1.80 Hz, H6'b); ^{13}C NMR [??, CD_3OD, 175 MHz): 101.40 (C1), 155.10 (C3), 108.92 (C4), 42.86 (C5), 75.45, (C6), 129.84 (C7), 151.50 (C8), 45.89 (C9), 61.73 (C10), 171.23 (C11), 100.40 (C1'), 74.98 (C2'), 78.52 (C3'), 71.61 (C4'), 77.81 (C5'), 62.84 (C6'); HRESI-MS (m/z) -ve mode 389.1074 $[M - H]^-$ (calcd for $C_{16}H_{21}O_{11}$: 389.1084), (m/z) +ve mode 413.1050 $[M + Na]^+$ (calcd for $C_{16}H_{22}O_{11}Na$: 413.1060). The spectra (IR, ^1H NMR, ^{13}C NMR and ^{13}C NMR DEPT) are given as supplementary figures 1–4. The structure of the compound was confirmed by comparison of the physical and spectroscopic data with those reported in the literature (Kim et al., 2005; Demirezer et al., 2006; Tzakou et al., 2007). To the best of our knowledge this is the first report for the occurrence of DAA from *Spermacoce hispida* and also an iridoid compound from this species (see Fig. 1).

3.2. In silico studies

DAA did not show any mutagenic, tumorigenic and irritant properties (Supplementary Table 2). DAA bound with the active site of PPARα at ASN'219/2HD2, GLU'282/OE1, LEU'331/O and TYR'334/HN amino acids like fenofibrate with comparable ligand efficiency (Supplementary Table 3). Hydrophobic interaction of DAA with PPARα is shown in Fig. 2.

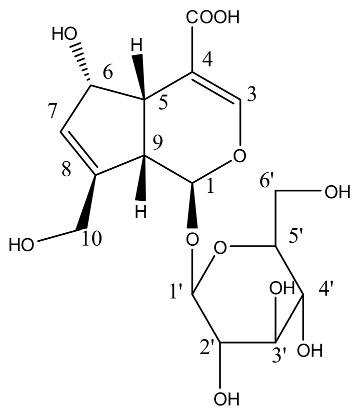

Fig. 1. Structure of deacetylasperulosidic acid (DAA).

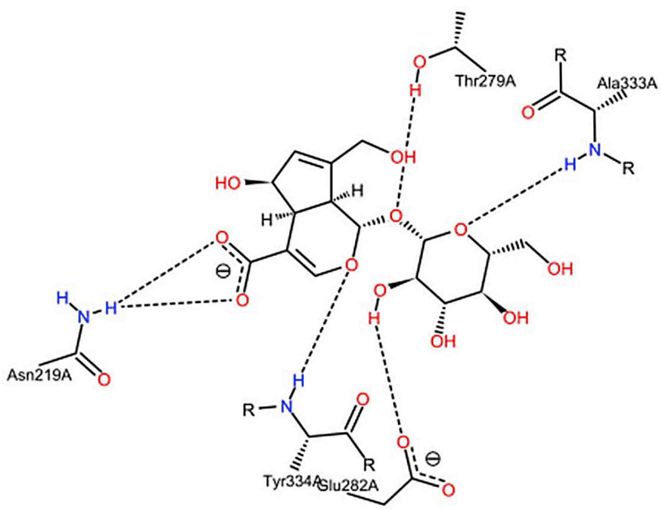

Fig. 2. Hydrophobic interaction of DAA with PPARα protein (PDB ID: 1I7G).

3.3. Effect of DAA on the viability of 3T3L1 preadipocytes

The treatment with DAA up to 100 μM concentration did not affect the viability of preadipocytes and the GI_{50} values were found as 609.28 μM. In matured adipocytes the treatment with DAA significantly lowered the viability from 20 μM concentration and GI_{50} was also low (145.08 μM) (Fig. 3).

3.4. Effect of DAA on PO-BSA induced steatosis in HepG2 cells

MTT assay showed that the GI_{50} and GI_{25} levels of DAA towards HepG2 cells as 542.30 and 281.33 μM, respectively; hence the doses for efficacy studies were fixed between 12.5 and 100 μM. Induction of steatosis with PO-BSA to HepG2 cells increased the lipotoxicity as indicated by the increase in the LDH, ALT and AST levels. The treatment with DAA significantly lowered the lipid accumulation as measured by the absorbance of ORO at 570 nm (Fig. 4) and by the reduction in the intracellular triglyceride content. The treatment with DAA significantly decreased the LDH leakage by 43.03% ($P < 0.05$) at 50 μM concentration. The treatment also reduced the leakage of ALT and AST by 50.48% and 42.53% at 50 μM concentration, respectively ($P < 0.05$) (Table 1).

3.5. Antihyperlipidemic effect of DAA in triton WR 1339 induced hyperlipidemia

The methanol soluble fraction of the aqueous extract of *Spermacoce hispida* seeds showed significant antihyperlipidemic effect in a dose dependent manner (Tables 2 and 3). Bioassay guided isolation of this fraction yielded DAA as the active ingredient (Supplementary Table 4). Treating the animals with DAA at 50 mg/kg concentration significantly lowered the TC, TG and LDL-c levels by 40.27, 46.00 and 63.65% respectively. The treatment also reduced the TC/HDL-c ratio in a significant manner. These effects were comparable with that of fenofibrate (Table 4).

3.6. Effect of DAA on anthropometric parameters of HFD fed rats

HFD feeding for two months caused 44.1% increase in the body mass and the there was no intergroup variation before the start of the experiment. After one month of treatment, the animals treated with fenofibrate and DAA showed a mild reduction in the body weight; but these reductions were not significant. The animals treated with DAA at 10 mg/kg and fenofibrate showed significant decrease in the BMI and ratio of AC to TC. The treatment did not alter the SRBG (Table 5). Similarly reduction in the reteroperitoneal adipose tissue mass was observed in the treated animals; the masses of spleen and kidneys were not affected in a significant manner (Table 6).

3.7. Effect of DAA on serum lipid profile of HFD fed rats

After a month of treatment with DAA to HFD fed animals, significant decrease in the serum TC, TG and LDL-c levels was observed. The effects of DAA at 10 mg/kg on the reduction of TC and the ratio between TC to HDL-c were comparable with that of fenofibrate (Table 7).

3.8. Effect of DAA on selected serum liver function enzymes

Increase in the transaminases, phosphatases and LDH levels were observed with HFD feeding. The treatment with DAA significantly lowered AST and ALT levels; it was also comparable with that of fenofibrate. At 10 mg/kg concentration, the treatment with DAA lowered the phosphatases levels. Though there was a reduction in LDH, it was not significant (Table 8).

3.9. Effect of DAA on the adipose tissue and liver histology of HFD fed animals

Histology of adipose tissue clearly showed the hypertrophy of adipocytes with HFD feeding; the adipose tissue from the animals treated with fenofibrate and DAA had smaller adipocytes (Fig. 5). Hepatic histology of treated animals clearly showed the reduction in steatosis and MNC infiltration. The cells in fenofibrate treated animals and DAA at the concentration of 10 mg/kg concentration were eosionophilic (Fig. 6).

4. Discussion

Lifestyle and dietary alterations are considered as major tools in the management of obesity; due to lack of adherence by the patients and insignificant outcome, there is a need for pharmacological interventions. Studies indicated that there is steady increase in the use of nutraceuticals and natural products for the control of obesity. In this study, we reported the active constituent from *Spermacoce hispida* seeds that have been used traditionally by the non-institutionally trained *Siddha* practitioners to treat obesity. Previously Sivaelango and coworkers (2012) documented the antihyperlipidemic effect of ethanol extract of *Spermacoce hispida* seeds in triton induced acute hyperlipidemic rats. Likewise Kaviarasan and his colleagues (2011) reported the antihyperlipidemic effect of flavonoid rich fraction of *Spermacoce hispida* seeds in HFD fed rats. To the best of our knowledge, this is the first report on the bioassay guided fractionation of antihyperlipidemic principle from *Spermacoce hispida* seeds and the occurrence of DAA in

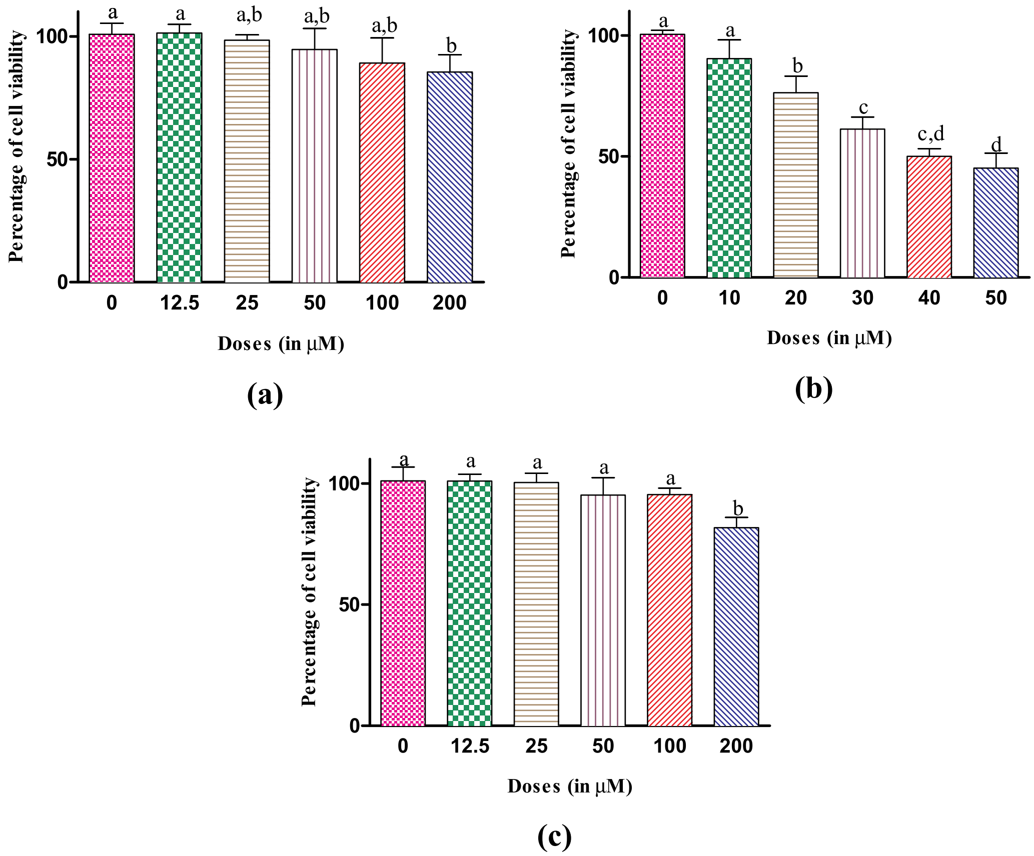

Fig. 3. Evaluation of the cytotoxicity of DAA in a) preadipocytes b) differentiated adipocytes and c) HepG2 cells. Cytotoxicity was assessed by MTT assay 24 h after treatment with DAA; Values indicate mean ± SD of four independent experiments; Values having same alphabet did not vary significantly (Tukey's HSD; $P \leq 0.05$).

Spermacoce hispida.

DAA was reported as one of the major iridoid glycosides from *Morinda citrifolia* L. (Noni) fruits (Potterat et al., 2007). The treatment with DAA was shown to reduce TNFα release from cultured mouse peritoneal macrophages (Li et al., 2006) and it inhibited LDL-c oxidation (Kim et al., 2005b). Treating the animals with DAA dose dependently increased the superoxide dismutase levels and decreased malondialdehyde concentrations (Ma et al., 2013). Health benefits of iridoids in reducing hyperlipidemia (Bai et al., 2009; Kojima et al., 2011; Liu et al., 2014; Zhou et al., 2015; Oi-Kano et al., 2017; Patel et al., 2018) were reported in the literature and most of them were seco-iridoids. The aglycones of the iridoids were reported to exert the anti-hyperlipidemic effect (Zhong et al., 2018; Oi-Kano et al., 2017). A study by Fujikawa and coworkers (2012) showed that asperuloside from *Eucommia ulmoides* exerted antiobesity effect by increasing the metabolic rate.

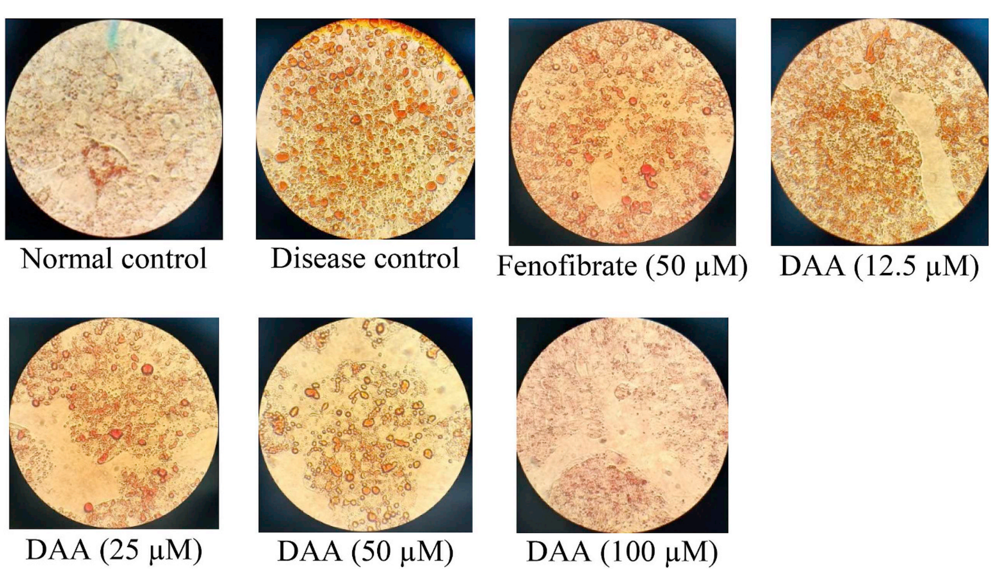

Fig. 4. Representative photomicrographs of DAA treated and ORO stained PO-BSA induced steatotic HepG2 cells 24 h after treatment. Photographs were taken at 400x.

Table 1
Effect of DAA isolated from the methanol extract of *Spermacoce hispida* seeds on the parameters of intracellular triglyceride accumulation and lipotoxicity of PO-BSA induced steatosis of HepG2 cells.

Groups	mg Tg/g Protein	%LDH	%ALT	%AST
Normal control	14.30 ± 0.85^{a,c}	11.07 ± 1.11^{a}	23.11 ± 4.15^{a}	18.17 ± 3.28^{a}
Disease control	29.86 ± 2.18^{b}	28.16 ± 1.52^{b}	40.51 ± 5.19^{b}	46.69 ± 3.52^{b}
Fenofibrate (50 µM)	15.50 ± 3.32^{a,c}	12.40 ± 0.67^{a,c}	25.04 ± 3.86^{a}	26.52 ± 8.28^{a}
DAA (12.5 µM)	19.53 ± 5.17^{c}	17.60 ± 4.19^{c}	39.84 ± 1.94^{b}	29.67 ± 7.74^{a}
DAA (25 µM)	12.13 ± 1.28^{a,c}	16.27 ± 1.20^{a,c}	25.14 ± 5.06^{a}	29.24 ± 5.22^{a}
DAA (50 µM)	12.41 ± 2.76^{a,c}	16.04 ± 1.85^{a,c}	20.06 ± 2.52^{a}	26.83 ± 3.64^{a}
DAA (100 µM)	9.81 ± 1.03^{a}	12.68 ± 0.56^{a,c}	18.38 ± 0.94^{a}	17.41 ± 3.70^{a}

Values indicate mean ± SD for triplicates; Values having same alphabet in a column did not vary significantly (Tukey's HSD, $P \leq 0.05$).

Table 2
Effect of the extracts of the *Spermacoce hispida* seeds on the lipid levels of Triton WR 1339 induced rats after 18 h of treatment at 200 mg/kg concentration.

Groups	Biochemical parameters				
	TC (mg/dL)	Tg (mg/dL)	HDL-c (mg/dL)	LDL-c (mg/dL)	TC/HDL-c
Negative Control	152.1 ± 12.61^{a}	300.3 ± 40.78^{a}	51.83 ± 12.49^{a}	40.22 ± 13.48^{a}	3.03 ± 0.54^{a}
Fenofibrate (100 mg/kg)	96.93 ± 9.33^{b}	188.3 ± 27.93^{b}	45.83 ± 5.63^{a}	13.44 ± 4.79^{b,c}	2.12 ± 0.15^{b,c}
Ethyl acetate soluble fraction	145.7 ± 12.16^{a}	297.0 ± 65.11^{a}	60.96 ± 9.59^{a}	25.36 ± 17.52^{a,c}	2.45 ± 0.54^{a,c}
Methanol soluble fraction	139.2 ± 14.00^{a}	284.4 ± 47.28^{a}	56.76 ± 9.60^{a}	25.56 ± 12.86^{a,c}	2.49 ± 0.32^{a,c}
Methanol insoluble fraction	94.76 ± 10.44^{b}	170.3 ± 19.26^{b}	50.38 ± 9.71^{a}	10.31 ± 4.99^{b,c}	1.91 ± 0.26^{b,c}

Values indicate mean ± SD for five animals; Values having same alphabet in a column did not vary significantly (Tukey's HSD, $P \leq 0.05$).

Table 3
Effect of methanol extract of *Spermacoce hispida* seeds on lipid levels of Triton WR 1339 induced rats after 18 h of treatment.

Groups	Biochemical parameters				
	TC	Tg	HDL-c	LDL-c	TC/HDL-c
Negative Control	161.1 ± 7.83^{a}	283.1 ± 15.80^{a}	64.29 ± 7.80^{a}	40.16 ± 9.83^{a}	2.52 ± 0.26^{a}
Fenofibrate (100 mg/kg)	90.02 ± 11.09^{b}	118.1 ± 25.17^{b}	59.34 ± 7.54^{a}	7.05 ± 1.47^{b}	1.51 ± 0.05^{b}
Methanol soluble fraction (100 mg/kg)	111.9 ± 10.46^{b}	174.4 ± 23.43^{c}	63.01 ± 7.16^{a}	13.98 ± 8.65^{b}	1.78 ± 0.12^{b}
Methanol soluble fraction (200 mg/kg)	90.23 ± 12.34^{b}	119.1 ± 16.21^{b}	58.99 ± 12.2^{a}	7.41 ± 2.47^{b}	1.54 ± 0.14^{b}

Values indicate mean ± SD for four animals; Values having same alphabet in a column did not vary significantly (Tukey's HSD, $P \leq 0.05$).

Table 4
Effect of DAA isolated from the methanol extract of *Spermacoce hispida* seeds on lipid levels of Triton WR 1339 induced rats after 18 hours of treatment.

Groups	TC	Tg	HDL-c	LDL-c	TC/HDL-c
Negative control	165.5 ± 16.21^{a}	250.2 ± 11.46^{a}	58.29 ± 6.08^{a}	57.17 ± 10.7^{a}	2.84 ± 0.13^{a}
Fenofibrate (50 mg/kg)	99.62 ± 15.08^{b}	136.9 ± 17.20^{b}	48.53 ± 6.91^{a}	23.71 ± 9.61^{b}	2.06 ± 0.29^{b}
DAA (25 mg/kg)	113.9 ± 14.83^{b}	150.3 ± 16.76^{b}	50.10 ± 6.76^{a}	33.77 ± 10.74^{b}	2.29 ± 0.31^{b}
DAA (50 mg/kg)	98.85 ± 12.74^{b}	135.1 ± 12.08^{b}	51.05 ± 1.75^{a}	20.78 ± 8.99^{b}	1.93 ± 0.19^{b}

Values indicate mean ± SD for four animals; Values having same alphabet in a column did not vary significantly (Tukey's HSD, $P \leq 0.05$).

Table 5
Effect of DAA isolated from the methanol extract of *Spermacoce hispida* seeds on the body weight of the HFD fed animals after one month of treatment.

Groups	Body weight (in g)		BMI	AC	AC/TC	SRBG
	Initial	Final				
HFD control	224.2 ± 07.98^{a}	247.2 ± 08.90^{a}	0.186 ± 0.005^{a}	10.50 ± 0.77^{a}	1.598 ± 0.06^{a}	4.207 ± 1.71^{a}
Fenofibrate (30 mg/kg)	228.2 ± 18.06^{a}	239.3 ± 10.73^{a}	0.132 ± 0.032^{b,c}	11.13 ± 1.55^{a}	1.252 ± 0.28^{b}	5.038 ± 5.16^{a}
DAA (5 mg/kg)	229.3 ± 15.28^{a}	237.7 ± 09.02^{a}	0.154 ± 0.017^{b}	10.17 ± 0.44^{a}	1.571 ± 0.13^{a}	5.795 ± 2.20^{a}
DAA (10 mg/kg)	223.7 ± 16.18^{a}	234.7 ± 12.55^{a}	0.120 ± 0.015^{c}	10.10 ± 1.17^{a}	1.229 ± 0.09^{b}	4.724 ± 1.35^{a}

Values indicate mean ± SD for six animals; BMI – Body Mass Index, AC – Abdominal circumference (in mm), AC/TC - Abdominal circumference/Thoracic circumference; SRBG – Specific Rate of Body Mass Gain (in g/kg); Values carrying same alphabets did not vary significantly (Tukey's HSD; $P \leq 0.05$).

Table 6
Effect of DAA isolated from the methanol extract of *Spermacoce hispida* seeds on relative organ weights of the HFD fed animals after one month of treatment.

Groups	Liver	Spleen	Reteroperitoneal adipose tissue	Kidneys
HFD control	3.112 ± 0.10[a]	0.263 ± 0.02[a]	2.264 ± 0.81[a]	0.736 ± 0.03[a]
Fenofibrate (30 mg/kg)	3.080 ± 0.56[a]	0.216 ± 0.03[a]	0.365 ± 0.19[b,c]	0.703 ± 0.05[a]
DAA (5 mg/kg)	2.869 ± 0.16[a]	0.217 ± 0.03[a]	1.054 ± 0.61[b]	0.744 ± 0.04[a]
DAA (10 mg/kg)	2.623 ± 0.32[a]	0.217 ± 0.05[a]	0.199 ± 0.11[c]	0.698 ± 0.02[a]

Values (in g/100 g body weight) indicate mean ± SD for six animals; Values carrying same alphabets did not vary significantly (Tukey's HSD; $P \leq 0.05$).

Table 7
Effect of DAA isolated from the methanol extract of *Spermacoce hispida* seeds on the lipid profile of the HFD fed animals after one month of treatment.

Groups	Total cholesterol	Triglycerides	HDL cholesterol	LDL cholesterol	TC/HDL-c
HFD control	76.09 ± 5.62[a]	130.3 ± 11.15[a]	34.10 ± 0.81[a]	15.93 ± 5.08[a]	2.230 ± 0.12[a]
Fenofibrate (30 mg/kg)	58.69 ± 5.25[b]	74.98 ± 11.15[c]	35.58 ± 4.95[a]	8.10 ± 4.65[c]	1.667 ± 0.19[c]
DAA (5 mg/kg)	62.69 ± 5.71[b]	103.2 ± 21.12[b]	31.60 ± 5.56[a]	12.12 ± 5.66[b,c]	2.106 ± 0.10[b]
DAA (10 mg/kg)	54.63 ± 6.22[b]	78.63 ± 13.58[c]	32.04 ± 1.17[a]	6.86 ± 3.70[b,c]	1.706 ± 0.18[c]

Values (in mg/dL., except TC/HDL-c) indicate mean ± SD for six animals; Values carrying same alphabets did not vary significantly (Tukey's HSD; $P \leq 0.05$).

Table 8
Effect of DAA isolated from the methanol extract of *Spermacoce hispida* seeds on selected liver function enzymes of HFD fed animals after one month of treatment.

Groups	AST	ALT	ALP	ACP	LDH
HFD control	182.2 ± 60.15[a]	170.5 ± 33.79[a]	22.63 ± 8.79[a]	6.00 ± 1.16[a]	621.3 ± 160.5[a]
Fenofibrate (30 mg/kg)	99.12 ± 9.19[b]	64.37 ± 6.85[c]	7.04 ± 4.27[b]	2.82 ± 0.89[b]	587.7 ± 162.1[a]
DAA (5 mg/kg)	119.2 ± 20.78[a,b]	119.2 ± 20.78[b]	13.24 ± 7.98[a,b]	5.33 ± 0.48[a]	566.5 ± 246.9[a]
DAA (10 mg/kg)	139.0 ± 22.51[b]	109.5 ± 33.48[b]	4.99 ± 1.79[b]	3.36 ± 0.71[b]	695.0 ± 53.51[a]

Values (in IU/L) indicate mean ± SD for six animals; Values carrying same alphabets did not vary significantly (Tukey's HSD; $P \leq 0.05$).

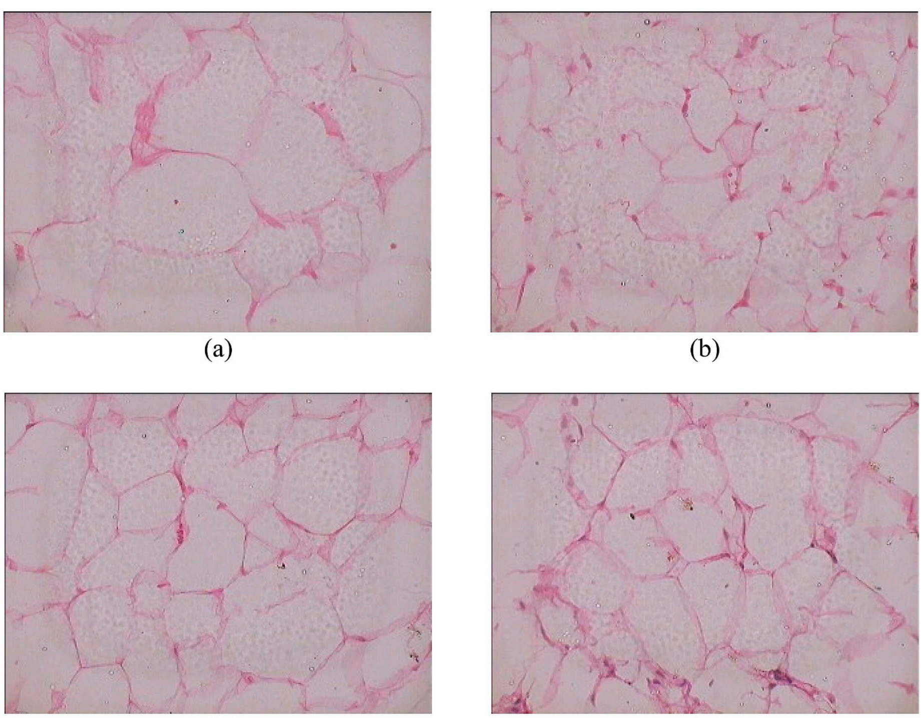
(a) (b) (c) (d)

Fig. 5. Representative histological images of reteroperitoneal adipose tissue sections from high fat diet fed rats after 28 days of treatment with DAA. Reteroperitoneal adipose tissues stored in buffered formalin were sectioned and stained with hematoxylin and eosine; the images were taken at 400x; **a)** HFD control; **b)** Fenofibrate (30 mg/kg); **c)** DAA (5 mg/kg) and **d)** DAA (10 mg/kg).

In silico prediction of the pharmacokinetic properties done in this study showed that DAA was safe and having low gastrointestinal absorption. It docked with PPARα with least binding energy and comparable ligand efficiency with fenofibrate. PPARα expresses majorly in the fat catabolizing tissues such as liver and muscle; its beneficial effect on hepatic steatosis was well established (Liss and Finck, 2017). Studies showed that PPAR$^{-/-}$ mice were prone to HFD induced adiposity and the reduction of adipocyte hypertrophy by the treatment with PPARα agonist (Stienstra et al., 2007). Docking analysis of DAA indicated that the compound might exert antihyperlipidemic effect through modulation of PPAR; however detailed investigations are needed to substantiate this claim.

In vitro experiments indicated that the treatment with DAA significantly lowered the viability of matured adipocytes, but not

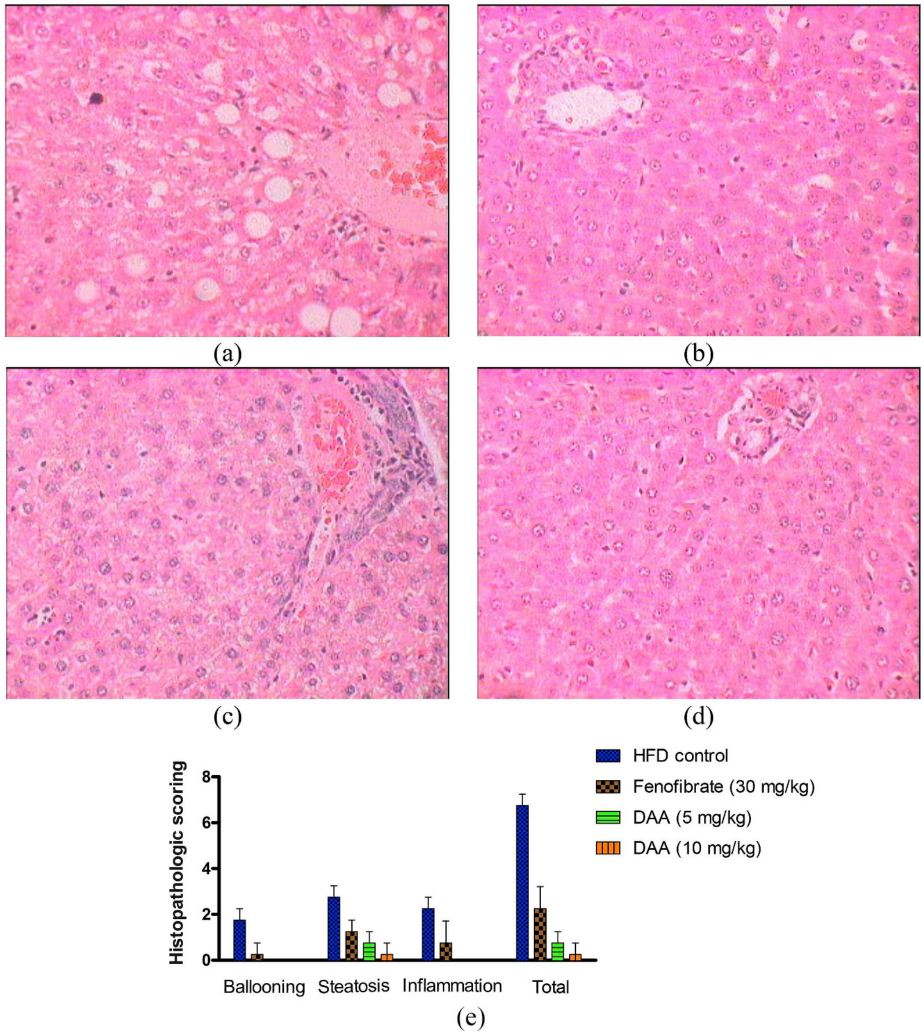

Fig. 6. Representative histological images of hepatic sections from high fat diet fed rats after 28 days of treatment with DAA. Livers stored in buffered formalin were sectioned and stained with hematoxylin and eosine; the images were taken at 400x; **a)** HFD control; **b)** Fenofibrate (30 mg/kg); **c)** DAA (5 mg/kg) and **d)** DAA (10 mg/kg); **e)** scoring of liver damage; histological sections were scored for hepatocyte ballooning (0–3), steatosis (0–3) and inflammation (0–3) according to the methodology of Itagaki et al. (2013).

preadipocytes. Adipose tissue mass is determined by the number and size of the adipocytes. Continuous deposit of triglycerides into the adipocytes causes hypertrophy of adipocytes; this leads to the formation of dysfunctional adipocytes that alter normal metabolism. Candidates that cause adipocyte apoptosis were suggested as a strategy to control obesity (Tung et al., 2017). The treatment with DAA also lowered lipid accumulation and lipotoxicity in steatotic HepG2 cells.

Triton WR 1339 induced hyperlipidemia is one of the widely used models to evaluate the acute antihyperlipidemic effect of the test materials. Systemic administration of triton WR 1339 causes a sharp increase in the serum lipid levels up to 24 h by blocking lipid uptake by peripheral tissues. The drugs that interfere on lipid biosynthesis exert antihyperlipidemic effect in this phase; while drugs that accelerate lipid removal exert antihyperlipidemic effect 24 h after triton WR 1339 administration (Zarzecki et al., 2014). In this experiment, DAA exerted antihyperlipidemic effect on the synthesis phase of triton WR 1339 induced hyperlipidemia.

Chronic feeding of HFD to rodents causes obesity, hypertriglyceridemia and hyperleptinemia; it mimics human obesity and metabolic syndrome (Buettner et al., 2007). Hence, we evaluated the antiobesity and chronic antihyperlipidemic effect of DAA using HFD fed animals. In this study, the treatment with DAA to HFD fed animals did not show any positive reductions in the body mass; this was in correlation with the short term observations made by Ma et al. (2013). The treatment significantly lowered the BMI, AC/TC ratio and reteroperitoneal adipose tissue mass. Interestingly the treatment did not affect the SRBG values; this indicated that the treatment specifically reduced the adipose tissue mass. Moderate hepatomegaly is one of the common clinical features of NAFLD (Basaranoglu and Neuschwander-Tetri, 2006) and it was also reported in HFD fed animals (Wat et al., 2009). The reduction in relative liver mass by DAA and fenofibrate indicated that their beneficial role on hepatomegaly.

The treatment with DAA lowered the serum total cholesterol, triglycerides and LDL-c levels significantly. Besides this, the treatment also lowered the transaminases and phosphatases levels in the serum of HFD fed animals. Previous clinical studies indicated that the treatment with fenofibrate significantly reduced transaminase levels in NAFLD subjects (Kostapanos et al., 2013). Fenofibrate was reported to decrease the cholesterol levels by increasing cholesterol efflux (Yaghoubi et al., 2017). The treatment with DAA also reduced the cholesterol level; its molecular mechanism is yet to be established. The histology of adipose tissue and liver indicated the reduction in adipocyte hypertrophy and improvement in steatosis as well as reduction of MNC infiltration, respectively. The treatment with DAA lowered the triglyceride accumulation in PO-BSA intoxicated HepG2 cells and improved the lipotoxicity. Previous report showed that DAA reduced the TNFα secretion from macrophages (Li et al., 2006); it indicates that the hepatoprotective efficacy of DAA might also be through modulating inflammatory pathways.

5. Conclusion

This study validated the traditional use of *Spermacoce hispida* seeds for the treatment of obesity; bioassay tailored fractionation of the aqueous extract of *Spermacoce hispida* yielded deacetyl asperulosidic acid (DAA) as the bioactive antihyperlipidemic principle. The treatment with DAA significantly lowered the viability of matured adipocytes and PO-BSA induced steatosis in HepG2 cells. In HFD fed animals, the treatment with DAA reduced BMI and the ratio of AC/TC without altering SRBG and body mass. Adipocyte hypertrophy and steatosis of liver cells were ameliorated by the treatment with DAA. This preliminary report had some limitations; the efficacy of DAA was not evaluated in primary cells and in genetic models of obesity. Further, the molecular mode of action of DAA was not clearly established. Further studies on the long term efficacy, safety and molecular mode of action of DAA are needed to reveal the antiobesity potential of *Spermacoce hispida* and deacetyl asperulosidic acid.

Author's contributions

PP and SI conceived the study and designed the experiments; SE, KBu, KB and OS did phytochemistry works; SN, TNS, KRN, SSM and TSK did the *in silico* and *in vitro* experiments; SE and SSD did the *in vivo* experiments; PP, SI, KB, OS and TNS processed the data and prepared the manuscript. All the authors have read and approved the manuscript before submission.

Conflicts of interest

The authors declare that they have no conflicts of interest.

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Appendix A. Supplementary data

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